Saiteja Venkateshwa Rao Dasari

Signal Integrity Engineer | Power Integrity | High-Speed Hardware

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SUMMARY

Electrical engineer with a B.E. in EEE and an M.S. from ASU (GPA 3.93), with hands-on signal and power integrity projects in Ansys HFSS 3D Layout and SIwave. Experienced in S-parameter extraction, crosstalk analysis, DDR4/HBM3 channel simulation, controlled-impedance design, and PDN modeling. Seeking Signal Integrity / Power Integrity roles.

EDUCATION

Arizona State University Aug 2024 – May 2026
M.S. Robotics and Autonomous Systems | GPA: 3.93/4.0 Tempe, AZ
Chaitanya Bharathi Institute of Technology Aug 2020 – May 2024
B.E. Electrical and Electronics Engineering Hyderabad, India
Relevant Coursework: Electromagnetic Fields, Signals and Systems, Analog Electronics, Digital Electronics, Power Electronics, Control Systems.

TECHNICAL SKILLS

SI/PI Analysis: S-parameter extraction (Touchstone), insertion/return loss, NEXT/FEXT crosstalk, eye diagrams, mask compliance, controlled impedance (single-ended/differential), differential vias, PDN impedance, decoupling.
Simulation Tools: Ansys HFSS 3D Layout, Ansys SIwave, Ansys Electronics Desktop, Ansys 2D/3D field solvers, IBIS models.
Interfaces: DDR4, HBM3, high-speed differential signaling.
Lab & Hardware: Oscilloscope, function generator, DMM, DAQ, bench power supply, schematic reading, PCB rework soldering, harness continuity testing.
Programming: Python (NumPy, Matplotlib), MATLAB, C, MS Excel.

EXPERIENCE

Arizona State University Jan 2025 – May 2026
Graduate Teaching Assistant, Instrumentation & Controls Lab and MATLAB Programming Tempe, AZ

- Supported 90+ students in instrumentation labs with oscilloscopes, function generators, DMMs, debugged flywheel measurement setups, wiring, and MATLAB Arduino experiments with grading.

Marut Drones (IIIT-Hyderabad) Aug 2023 – Mar 2024
Production Engineering Intern, UAV Systems Integration Hyderabad, India

- Assembled and verified UAV power distribution and avionics (ESCs, flight controllers, GPS, wiring harnesses); ran continuity checks, crimped and soldered connectors, and reworked PCB joints, raising production efficiency by 15%.
- Read schematics and wiring drawings to root-cause hardware bring-up faults; tuned ArduPilot parameters from flight logs across 3 UAV variants, improving flight stability by ~10%.
- Traced and fixed signal faults on PCBs, flight controllers, ICs, and radar modules using oscilloscopes and multimeters, restoring clean communication between onboard electronics and reducing EMI interface to other signal wires.

PROJECTS

HBM3 Channel and PDN Co-Simulation on a 2.5D Silicon Interposer Ansys HFSS 3D Layout, SIwave

- Modeled a 4-pair HBM3 differential channel on a 2.5D silicon interposer in HFSS 3D Layout and extracted its 16-port S-parameters, achieving −2.4 dB insertion loss and NEXT/FEXT under −39 dB at 3.2 GHz Nyquist for 6.4 Gbps NRZ signaling.
- Ran a 4-point parametric crosstalk study varying trace spacing from 8 to 15 µm while holding IL/RL constant to isolate coupling; quantified ~1 dB NEXT/FEXT reduction per µm and derived the trade-off between routing density and SI margin.
- Built a coupled power/ground-plane PDN model in SIwave from 1 MHz to 2 GHz, achieving a 2 mΩ impedance minimum at the 95 MHz decap SRF versus a 9.2 mΩ target (4.5× margin); a decap-count sensitivity sweep showed impedance degrading 3× when 50% of decaps were removed.

DDR4-2133 Channel Signal Integrity Analysis Ansys SIwave, Electronics Desktop, IBIS, S-parameters

- Ran SI analysis of 2133 MT/s DDR4 channels using IBIS-driven transient simulation in Ansys SIwave and Electronics Desktop, with extracted S-parameter models of the channel.
- Evaluated the resulting eye diagrams and confirmed mask compliance against JEDEC DDR4 requirements.

Controlled-Impedance Stackup Planning and Differential Via Optimization Ansys 2D/3D field solvers

- Designed microstrip and stripline interconnects on a multilayer PCB stackup to hit 40/50/60 Ω single-ended and 90/100/110 Ω differential impedance targets.
- Analyzed how stackup variations change characteristic impedance, and refined differential via transitions using the 3D field solver.

CERTIFICATIONS

Transmission Lines 101 – Dr. Eric Bogatin, The EEcosystem Electronics Masterclass
Essential Principles of Signal Integrity (EPSI) – Dr. Eric Bogatin, Teledyne LeCroy